Project Check-in 3

Sketch-to-Photo Synthesis with ControlNet-Guided Diffusion

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Team deliverable

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1. Repository and invitations

All project code and our check-in history are in one GitHub repository: https://github.com/LRozay/Sketch-to-Photo. The instructor (bertin.cordovadiba@sjsu.edu) and the TA (charansai.gandham@sjsu.edu) have been invited to it. The commits from September 27 to October 11, 2026 were reported in Check-ins 1 and 2. This check-in covers the fourteen commits made from October 14 to November 1, 2026, listed in Section 10. Section 3 shows how those commits complete the remaining work that Check-in 2 set out. This document is the updated draft of the final report. It contains the tentative abstract, the report outline, the literature survey, the approach, the models explored with the results we can already report, and the work left for the final report.

2. Finalized specifications

The scope has not changed since the proposal. Given a sketch and a short text prompt, we generate a photorealistic image. The baseline is Stable Diffusion 1.5 with the pretrained ControlNet canny and scribble checkpoints, loaded through Hugging Face Diffusers. We then adapt ControlNet with LoRA on a small set of real hand-drawn sketches while Stable Diffusion 1.5 stays frozen. We compare the pretrained baseline with the adapted model on the same sketches, and we ablate the ControlNet conditioning scale and the classifier-free guidance scale. Images are 512 × 512. Quality is measured with FID against held-out COCO photos, CLIP score for prompt agreement, and a small pairwise human survey. Training and batch inference run on the College of Engineering HPC, because latent diffusion at this resolution needs a GPU.

3. Progress on the Check-in 2 remaining work

Check-in 2 ended with five next steps: finish the synthetic and hand-drawn sets, run the baseline end to end on the HPC, record baseline FID and CLIP score, start the first LoRA run, and fill the results and discussion sections. Table 1 maps each step to the commits that completed it.

Table 1. Check-in 2 remaining work and the commits that complete it.

Check-in 2 step	Commits	What is now done
Finish the synthetic and hand-drawn sets	a340c71, 459da4d, 4be4f4a, 6d567b1	Canny and HED map generation; COCO subset selection (3,000 images by default, limited to 2,000 to 5,000) with captions exported as prompts; a held-out split of 500 COCO photos for FID; a collection manifest and progress report for the hand-drawn set, with the first 10 sketches and prompts planned.
Run the baseline end to end on the HPC	11dc4a7, 6bef738	Batch scribble inference over every complete hand-drawn triple and a SLURM job (1 GPU, 32 GB, 4 hours) that checks the data and runs the batch with seed 261.
Record baseline FID and CLIP score	b08d14f, 1638f39	CLIP score, FID, and survey-sheet commands, plus a recorder that writes the scores to results/baseline_metrics.json.
Start the first LoRA run	5b58b4f, 7930f83, 95ffa41	A SLURM job (1 GPU, 48 GB, 8 hours) for the rank-8, 500-step run; a fixed-seed baseline versus LoRA comparison on the same sketches; and the nine-cell scale grid repeated with the adapter.
Fill the results and discussion sections	f4d2fd4, 6db4ef3	The pairwise survey protocol, the expected failure cases, and the results and discussion outline, which are drafted in Sections 8 and 9 of this document.

Every step in Check-in 2 now has working code and an HPC job in the repository. The measured scores are not in yet. results/baseline_metrics.json is set to null on purpose until the jobs finish, and we do not report numbers that have not been measured. Running the two jobs on the completed hand-drawn set is the first item in Section 11.

4. Tentative abstract

Rough sketches are a natural way to describe a picture early in design, but pretrained ControlNet models work best on clean edge maps. Real freehand drawings have broken strokes, uneven lines, and missing detail. This project asks how well a Stable Diffusion 1.5 plus ControlNet pipeline transfers from synthetic edge maps to real hand-drawn sketches when only 50 to 100 real sketches are available for adaptation.

We pair a COCO 2017 subset of about 2,000 to 5,000 captioned photos with Canny and HED edge maps and use the captions as prompts, holding out part of the subset as the real set for FID. A separate set of hand-drawn sketches, each with a prompt and a target photo, is used to fine-tune and test. Stable Diffusion 1.5 stays frozen, and a rank-8 LoRA adapter with 731,136 trainable parameters, about 0.2% of ControlNet, is trained on the ControlNet attention projections. We compare the pretrained canny and scribble baselines against the adapted model using FID, CLIP score, and a pairwise human survey, and we sweep the conditioning and guidance scales. In early trials the pretrained model followed clean synthetic edges more reliably than incomplete hand drawings, and that gap is what the adapter is meant to close. We expect fine-tuning to improve how closely the image follows the sketch without hurting realism, and we will report where it still fails.

5. Outline of the final report

Table 2. Planned sections of the final report and their status in this draft.

Section	Status
1. Abstract	Drafted (Section 4)
2. Introduction and motivation	Outlined; will expand the motivation in the proposal and Section 4
3. Related work	Full survey in Check-in 2, summarized in Section 6
4. Data	Preparation code complete (Section 7); hand-drawn collection in progress
5. Method	Drafted (Section 7)
6. Experiments and evaluation setup	Drafted (Section 8); scoring code and HPC jobs complete
7. Results	Analysis and pre-run checks drafted (Section 8); scores pending the HPC runs
8. Discussion and failure cases	Drafted from the committed outline (Section 9)
9. Conclusion and future work	Not started; written after the measurements
10. References	Drafted (Section 12)

6. Literature survey (summary)

The full survey is in Section 5 of Check-in 2 and will carry into the final report. In short, the pipeline sits on four ideas. Ho et al. (2020) define diffusion models: a fixed forward process adds Gaussian noise, and a network learns to predict that noise so it can be removed step by step. Our paper reading assignment covered this paper, and its noise-prediction loss is the same loss our LoRA script uses. Rombach et al. (2022) run diffusion in the latent space of an autoencoder, so a 512 × 512 image becomes a 4 × 64 × 64 latent, which is what makes Stable Diffusion 1.5 cheap enough to fine-tune around.

Zhang et al. (2023) add spatial control to a locked model through ControlNet, using a trainable copy of the encoder joined to the locked U-Net by zero-initialized convolutions, so training starts without disturbing the pretrained model. Hu et al. (2022) introduce LoRA, which adds a low-rank update W + BA to chosen weight matrices so only a small number of new parameters are trained. We apply it to the ControlNet attention projections rather than to the full network.

Canny (1986) and HED (Xie and Tu, 2015) supply the synthetic sketches. FID (Heusel et al., 2017) compares Inception feature statistics of real and generated images, and CLIP score (Radford et al., 2021) is the cosine similarity between image and prompt embeddings. COCO (Lin et al., 2014) provides the photos and captions. We also considered pix2pix (Isola et al., 2017), a conditional GAN that learns paired translation from scratch. It needs a large paired set and has no frozen text-to-image prior, so we do not pursue it.

7. Approach

The pipeline is implemented in src/, with HPC jobs in hpc/ and evaluation notes in evaluation/. It has six stages.

COCO data and synthetic sketches. src/prepare_coco_subset.py samples a COCO 2017 subset with seed 261 (3,000 images by default; the script refuses sizes outside 2,000 to 5,000), resizes each photo to 512 × 512, and writes its first caption as the prompt. src/split_coco_for_fid.py holds out 500 of those photos as the real set for FID and keeps the rest for pairing. src/prepare_synthetic_edges.py writes a three-channel Canny map, with

thresholds 100 and 200 by default, or an HED map produced with OpenCV's DNN module. The HED weights are passed in at run time and are not stored in the repository.

Hand-drawn data. Each hand-drawn example is a target photo, a sketch, and a one-line prompt, matched by a shared filename stem across images/, sketches/, and prompts/. data/hand_drawn/collection_manifest.csv lists the planned sketches with their category and draft prompt, and src/report_hand_drawn_progress.py counts complete triples against the 50 to 100 target. src/check_hand_drawn_data.py is the preflight check that every HPC job runs first.

Pretrained baseline. Stable Diffusion 1.5 and one ControlNet (canny or scribble) generate a 512 × 512 image with the UniPC sampler, 30 steps, guidance scale 7.5, conditioning scale 1.0, and seed 261 by default. Every run saves the photo, the conditioning image ControlNet saw, and a text record of its settings. src/run_batch_baseline.py runs this over every complete triple and can attach a LoRA adapter.

LoRA adaptation. The text encoder, autoencoder, and U-Net are frozen. LoRA with rank 8, alpha 8, and Gaussian initialization is attached to the ControlNet projections to_q, to_k, to_v, and to_out.0. A training step encodes the photo to a latent, samples a timestep, adds DDPM noise, and conditions the U-Net on the sketch through ControlNet. The loss is mean squared error between the predicted and true noise. We use AdamW with learning rate 1e-4, batch size 1, and 500 steps by default.

Comparison and scale grid. src/compare_baseline_lora.py renders the same sketches with the pretrained ControlNet and with the adapter into separate folders, with the seed, scales, and step count fixed so a pair differs only by the model. src/run_baseline_ablation.py sweeps conditioning scales 0.5, 1.0, 1.5 and guidance scales 5.0, 7.5, 12.0 and now accepts --lora-path, so the same nine-cell grid is produced for the baseline and for the adapter.

Evaluation and HPC jobs. src/evaluate_generations.py reports the mean CLIP ViT-B/32 cosine similarity between each image and its prompt, computes FID from 2048-dimensional Inception-v3 features, and writes the blank survey sheet. src/record_baseline_metrics.py saves the baseline scores to results/baseline_metrics.json. evaluation/survey_protocol.md fixes the rating rule: for each sketch, raters mark baseline, lora, or tie for the photo that follows the sketch more closely, and judge sketch following before photorealism. hpc/run_baseline.slurm and hpc/run_lora_train.slurm run the baseline batch and the first LoRA training on one GPU node each.

Full U-Net fine-tuning and a pix2pix model were set aside because a small sketch set cannot support training a photorealistic generator from scratch.

8. Experiments and results

8.1 Models explored

We have explored two models: the pretrained canny ControlNet and the pretrained scribble ControlNet, both on a frozen Stable Diffusion 1.5. Both run through the baseline and sweep scripts, and the scribble model also runs in batch over the hand-drawn set. The third configuration, the scribble ControlNet with a rank-8 LoRA adapter, is the model we are developing. Its training script, HPC job, comparison script, and adapter grid are complete. The training script prints the loss at step 1, every 50 steps, and the final step.

8.2 Analytical results: LoRA parameter budget

The size of the adapter follows directly from the architecture. The SD 1.5 ControlNet copies the U-Net encoder. It has seven transformer blocks: two at width 320, two at 640, two at 1280, and one in the middle block at 1280. Each block has a self-attention layer, where to_q, to_k, to_v, and to_out.0 all map d to d, and a cross-attention layer, where to_k and to_v map the 768-dimensional CLIP text embedding to d. LoRA of rank r on a layer with input size m and output size n adds r(m + n) parameters, so one block adds r(14d + 1536). Across the 56 adapted

layers, rank 8 gives 731,136 trainable parameters, about 0.2% of the roughly 361 million parameters in ControlNet. Because alpha equals the rank, the LoRA scale alpha/r is 1.

Table 3. Trainable LoRA parameters at rank 8, by ControlNet block.

Width d	Transformer blocks	LoRA parameters
320	2 (down block 1)	96,256
640	2 (down block 2)	167,936
1280	2 (down block 3)	311,296
1280	1 (mid block)	155,648
Total	7 blocks, 56 adapted layers	731,136

The count is linear in the rank, so rank 4 would train 365,568 parameters, rank 16 would train 1,462,272, and rank 32 would train 2,924,544. With 50 to 100 sketches, the default 500 steps at batch size 1 show each sketch about 5 to 10 times. These two numbers are the levers we will use if the adapter underfits or starts copying strokes.

8.3 Observations from the baseline trials

The observations below come from the Check-in 1 trials with the pretrained baseline.

Canny threshold tradeoff. Lower thresholds keep thin and broken contours but also keep noise. Higher thresholds give a cleaner map but drop weak detail. This matters because the thresholds decide how close a synthetic sketch is to a real drawing.

HED versus Canny. HED maps follow object boundaries more continuously than Canny, but they sometimes keep texture and background edges.

Baseline gap. The pretrained ControlNet followed clean synthetic edges more reliably than incomplete hand drawings. This is the main reason we adapt ControlNet with LoRA instead of only tuning guidance.

8.4 Pipeline checks before the HPC runs

Before spending GPU time, we reviewed the scripts against the method. One issue from our review, the scale grid not accepting an adapter, is already fixed in commit 95ffa41. Three issues remain and will be fixed and committed before the first scored run.

FID square root. evaluate_generations.py takes the matrix square root of the product of the real and generated covariance matrices with torch.linalg.eigh. That routine assumes a symmetric matrix, and the product of two covariance matrices is generally not symmetric. In a check on synthetic Gaussian features (500 samples, 64 dimensions), the script's formula gave 168.70 while scipy.linalg.sqrtm gave 207.96, so the current code would underestimate FID by about 19%. The fix is to use the symmetric form, the square root of Σr^½ Σg Σr^½, which has the same trace, and to confirm the result against a standard FID package. The torchvision Inception weights also differ from the ones used for published FID, so our FID is only comparable within this project.

Scribble polarity. The pretrained scribble checkpoint was trained on white strokes on a black background, while drawings on paper are dark strokes on white. Neither the inference script nor the training script inverts the input, so hand-drawn sketches must be thresholded and inverted before they reach ControlNet.

FID data wiring. record_baseline_metrics.py compares a generated COCO folder with data/coco/images, the whole subset, rather than with the 500 held-out photos in fid_real.txt. The real folder will be built from that split, and the COCO edge maps will be run through the batch baseline to produce the generated set for FID.

8.5 Evaluation matrix

Table 4 shows what each configuration will be scored on. FID is computed on the larger COCO-based generation set against the held-out COCO photos, because FID on 50 to 100 images is too noisy. CLIP score is the mean over prompts. The survey compares the scribble baseline with the adapted model.

Table 4. Configurations, status, and the metrics each will report.

Configuration	Status	Metrics to report
Pretrained canny	Inference and sweep complete	FID, CLIP score
Pretrained scribble	Batch inference and HPC job complete; scores pending the run	FID, CLIP score, survey (baseline side)
Scribble + LoRA (rank 8, 500 steps)	Training job, comparison, and adapter grid complete; training run pending	FID, CLIP score, survey (adapted side)

9. Discussion and failure cases

evaluation/expected_failure_cases.md records three failure types from the Check-in 1 trials and the literature: thin or broken strokes that the baseline ignores, extra objects that guidance invents to satisfy the prompt, and copied stroke marks after fine-tuning on a small set. The first was already seen in Check-in 1. The final side-by-side grids will keep at least one example of each. Two further risks come from Section 8. Each sketch is seen 5 to 10 times during training, so stroke copying is a real possibility, and an uninverted sketch would give the scribble model an almost empty condition, which would look like a model failure but is a preprocessing error.

The claim we will test is that LoRA improves sketch following on messy strokes without fully solving conflicts between the prompt and the sketch. We will not state a conclusion until the comparison images and survey votes exist.

10. Meaningful check-ins

Repository: https://github.com/LRozay/Sketch-to-Photo

Table 5 lists the fourteen commits on the main branch from October 14 to November 1, 2026, above the 8 to 10 expected for a three-person team. The earlier commits, from September 27 to October 11, were reported in Check-ins 1 and 2 and are not repeated here.

Table 5. Commits on the main branch, October 14 to November 1, 2026.

Date	Commit	What it added
Oct 14, 2026	a340c71	Canny and HED synthetic sketch generation for the COCO pairs
Oct 19, 2026	b08d14f	CLIP score, FID, and the pairwise survey sheet
Oct 21, 2026	980dd27	The Check-in 2 report
Oct 23, 2026	459da4d	COCO subset preparation and caption prompt export
Oct 24, 2026	4be4f4a	Held-out COCO photograph split for FID scoring
Oct 24, 2026	6d567b1	Hand-drawn collection manifest and pairing progress report
Oct 26, 2026	11dc4a7	Batch scribble baseline inference over the hand-drawn set
Oct 27, 2026	6bef738	HPC job script for the end-to-end pretrained scribble baseline
Oct 28, 2026	1638f39	Baseline FID and CLIP score recording under results/
Oct 28, 2026	7930f83	Fixed-seed baseline versus LoRA comparison on the same sketches

Date	Commit	What it added
Oct 29, 2026	95ffa41	Nine-cell scale grid repeated for the LoRA adapter
Oct 30, 2026	5b58b4f	HPC job script to start the first rank-8 LoRA run
Oct 31, 2026	f4d2fd4	Pairwise survey protocol and expected failure-case notes
Nov 1, 2026	6db4ef3	Results and discussion outline sections, with no invented scores

11. Remaining work

The work left for the final report, in order, is to fix the three issues in Section 8.4, finish drawing the hand-drawn set from the 10 planned sketches up to 50 to 100 complete triples, submit the baseline and LoRA jobs on the HPC, record FID and CLIP score for all three configurations, run the fixed-seed comparison and both nine-cell grids, collect the survey votes, and write the introduction, results, discussion, and conclusion. Leo Rozay owns data preparation, Sriyavarma Saripella owns the ControlNet and LoRA code, and Venkata Satya Charan Naga Deep Patrini owns the metrics, the survey, and the report.

12. References

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